

XIAMEN UNIVERSITY MALAYSIA

厦門大學馬來西亞分校

COM105 Engineering Physics (I)

Lecture 2: Kinematics - Motion Along A Straight Line (Motion in 1D)

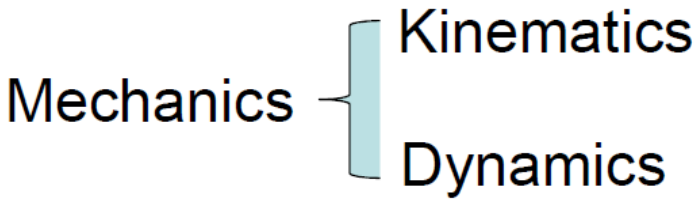
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In this lecture, you'll learn…

- how the ideas of displacement and average velocity help us describe straight-line motion.
- the meaning of instantaneous velocity; the difference between velocity and speed.
- how to use average acceleration and instantaneous acceleration to describe changes in velocity.
- how to solve problems in which an object is falling freely under the influence of gravity alone.
- how to analyze straight-line motion when the acceleration is not constant.

Mechanics, Kinematics and Dynamics

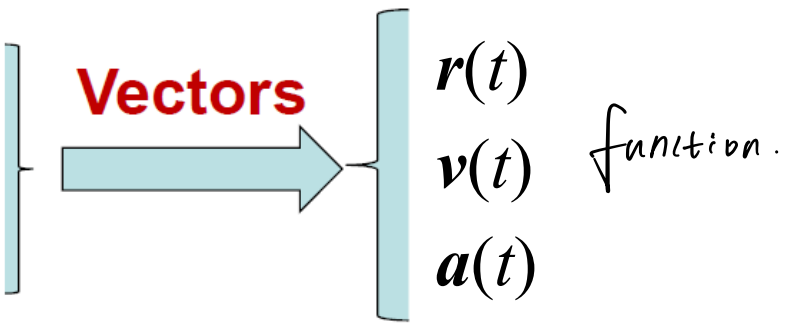
力学 运动学 动力学



- Mechanics
  - The relationship among force, matter and motion, including kinematics and dynamics.
- Kinematics
  - Describe the motion of the matter in terms of space and time
  - The agents that cause the motion are not considered
- Dynamics
  - The relationship between the motion and its causes

Physical Quantities Studied in Kinematics

- Kinematics: describe the motion of the matter in terms of **space (*r*)** and **time** (t) 位置矢量
  - Space: position vector ***r***, vector
  - Time: t, scalar
- Physical quantities studied in kinematics
  - Displacement 位移
  - Velocity v.
  - Acceleration a



Motion in a Straight Line (1D Motion)

- The simplest motion
- Example: Swimming race, free falling objects





直线

Motion in a Straight Line (1D Motion)

近似质点

Particle Approximation is used in the study

- The moving object is assumed as a particle to simplify the study.

粒子/质子



- A point on the dragster, such as its front end, represents the entire dragster by that point as a particle in the study of motion in a straight line.

Physical Quantities in 1D Motion

- Motion in a straight line, or 1D motion



固定

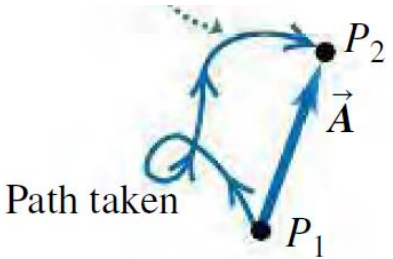
  - direction is determined/fixed (for example, x-direction) 坐标系固定.
  - only one component (for example, x-component) 只有一个分量.
- For the physical quantities in 1D Motion, we just need to study their magnitude, *r(t)*, *v(t)*, and *a(t)*. They are all scalar (the simplest motion)

一维运动只用研究标量

Three scalar physical quantities we need to study 1D Motion

Displacement in 1D Motion

- Displacement, as defined in Chapter 1, is a vector 矢量
  - A displacement is always a straight arrow directed from the starting position to the ending position. It does not depend on the path taken, even if the path is curved. adj. 弯曲的

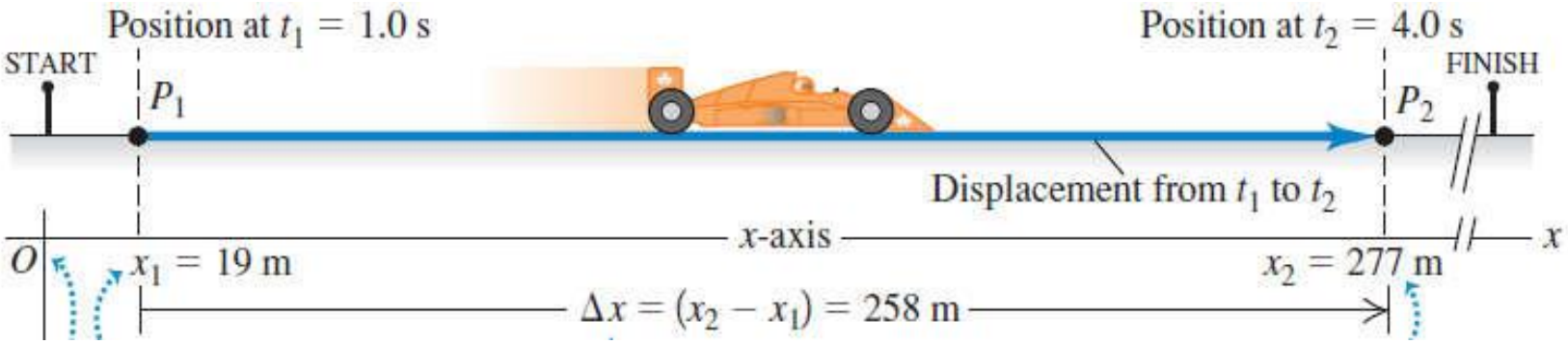


- When an object/particle moves in a straight line, i.e. 1D motion, its direction can be assumed in the x-coordinate direction.
  - Initial position x_1, final position x_2, then the magnitude of the displacement is: $\Delta x = x_2 - x_1$

    where Δx is scalar 标量

Displacement in 1D Motion

Example: Dragster's 1D motion



- The motion of a particle is completely known if its position is known at all times.

Displacement in 1D Motion

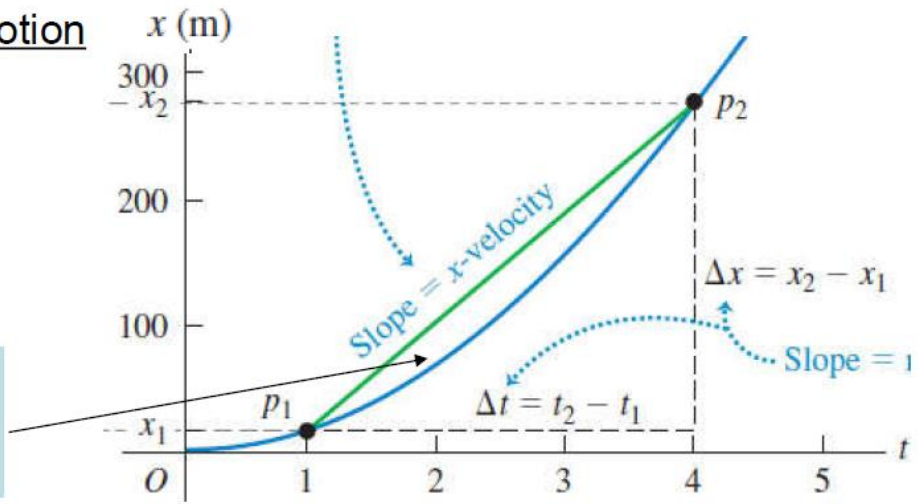
- Position-time graph: plot of *x(t)* against *t* for 1D motion

x(t) : position of the particle/object

Displacement for 1D motion: change of particle's position, Δx

$$\Delta x = x(t_2) - x(t_1) \quad \text{or} \quad \Delta x = x_2 - x_1 \qquad (2.1)$$

Example: Dragster's 1D motion



$x(t)$ vs. t : the blue curve in the figure

Velocity

- Average Velocity 平均速度
  - The average velocity is the displacement or the change in position of a particle divided by the time interval during which the change occurs.

$$v_{\text{av-}x} = \frac{\Delta x}{\Delta t} = \frac{x_2 - x_1}{t_2 - t_1} \ (\text{m/s.}) \qquad (2.2)$$

  - **Velocity is a vector, speed is a scalar**

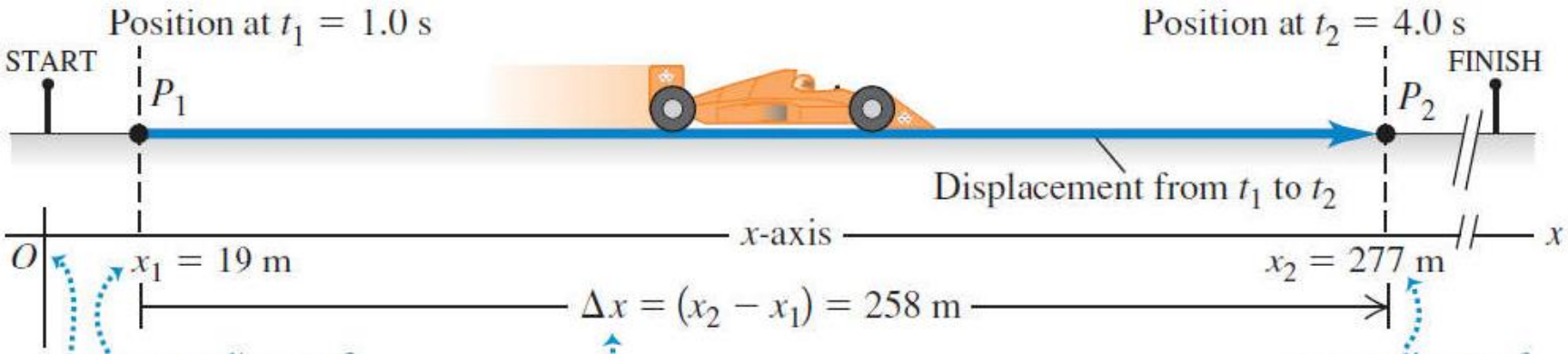
向量

Speed is the magnitude of velocity

Velocity

Example: Dragster's 1D motion

2.1 Positions of a dragster at two times during its run.



$$v_{\text{av-}x} = \frac{\Delta x}{\Delta t} = \frac{258 \text{ m}}{3.0 \text{ s}} = 86 \text{ m/s}$$

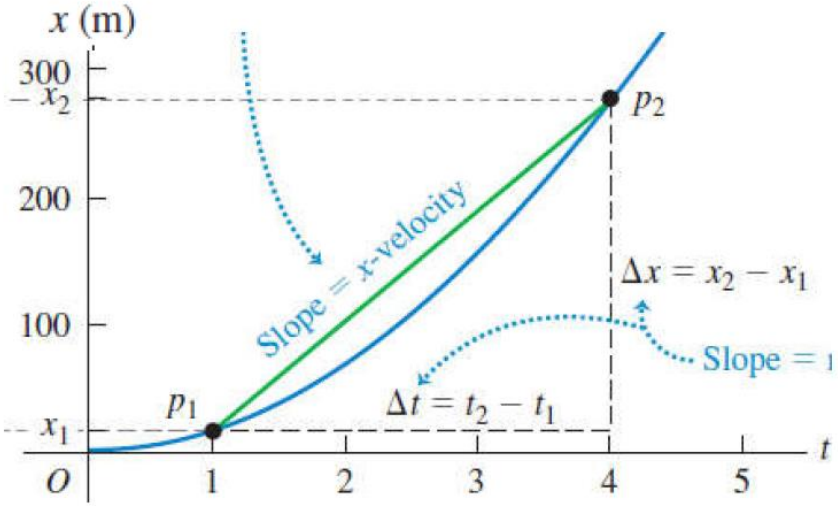
only have 2 point

Velocity

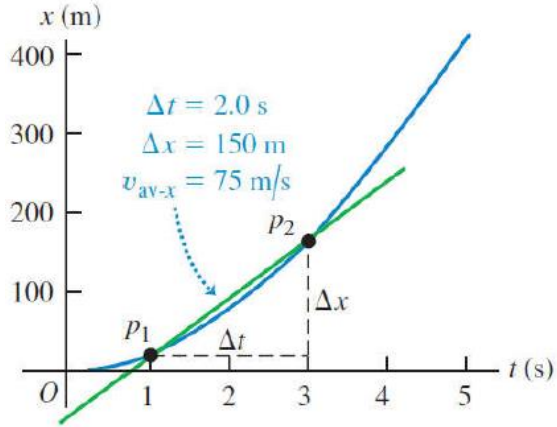
- For a displacement along the x-axis, an object's average x-velocity $v_{av\text{-}x}$ equals the slope of a line connecting the corresponding points on a graph of position (x) versus time (t).

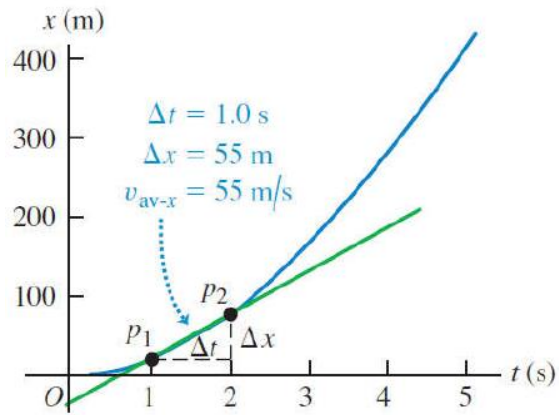
Example: Dragster's 1D motion: from the plot of $x(t)$-t

$t_1 = 1.0\text{s}, x_1 = 19\text{m},$
$t_2 = 4.0\text{s}, x_2 = 277\text{m}$



$$v_{av\text{-}x} = \frac{\Delta x}{\Delta t} = \frac{277\text{ m} - 19\text{ m}}{4.0\text{ s} - 1.0\text{ s}} = \frac{258\text{ m}}{3.0\text{ s}} = 86\text{ m/s}$$





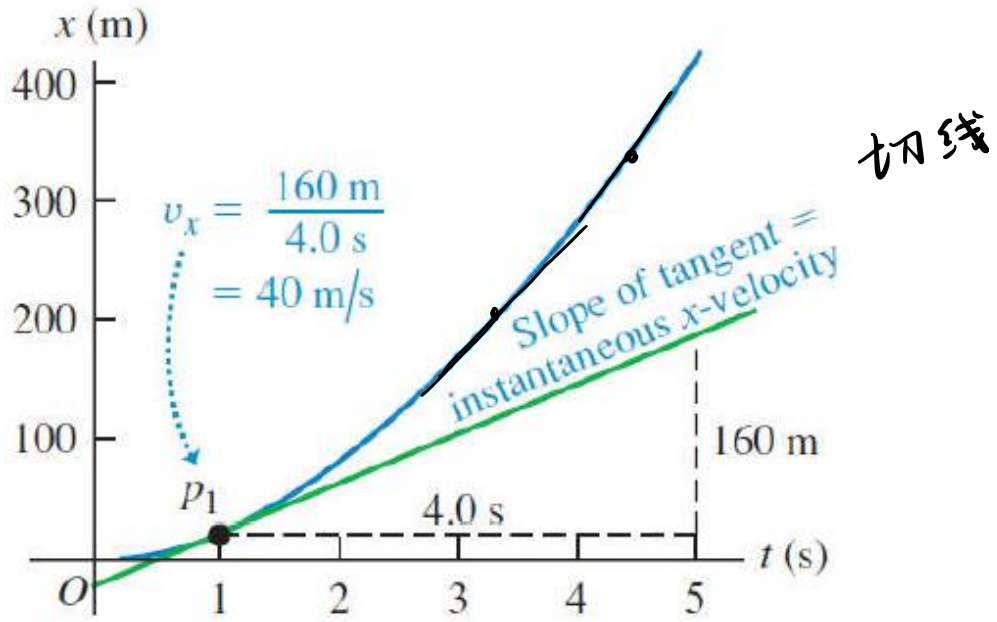
Velocity

- Instantaneous Velocity
  - The instantaneous velocity is the limit of the average velocity as the time interval approaches zero; it equals the instantaneous rate of change of particle's *x*-coordinate, or the derivative of the position *x* with respective to time *t*.

$$v_x = \lim_{\Delta t \to 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt} \text{ (m/s)} \quad (2.3)$$

Velocity

- Instantaneous Velocity from the plot of *x(t)*-*t*



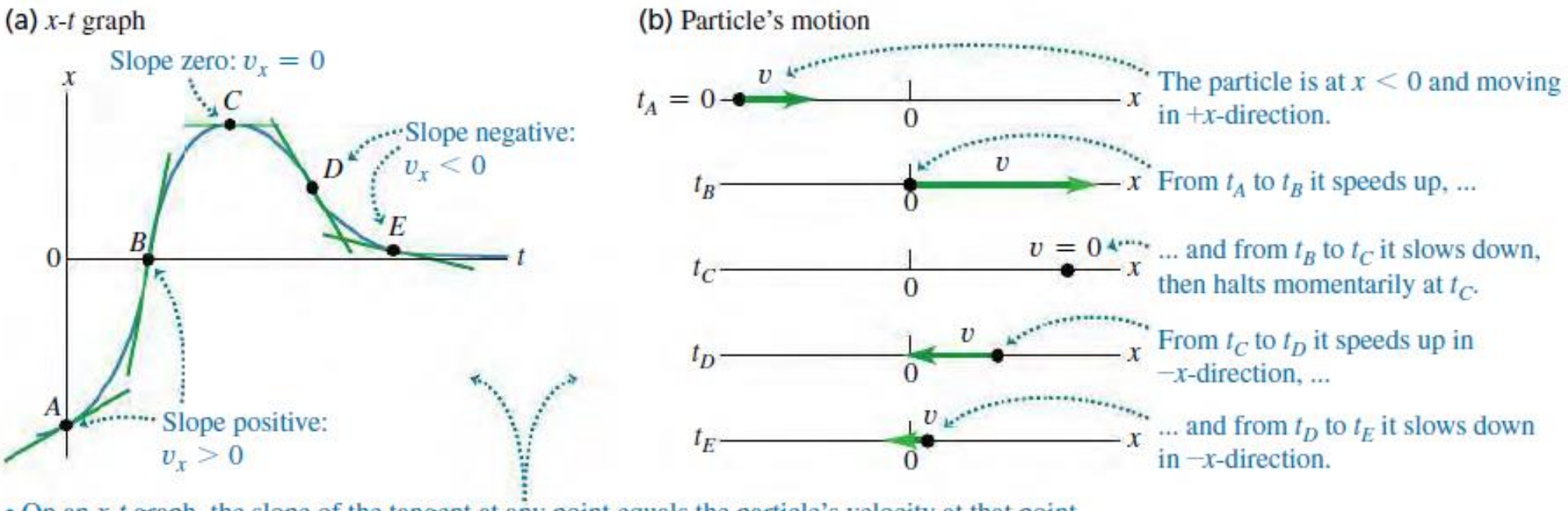
- The instantaneous velocity at any point is equal to the slope of the tangent to the $x(t)$-t curve at that point.

Finding the equation of a tangent to a curve: https://www.youtube.com/watch?v=AUbTx0IdTIw

Velocity

(a) The *x*-*t* graph of the motion of a particular particle.

(b) A motion diagram showing the position and velocity of the particle at each of the times labeled on the *x*-*t* graph.



- On an *x*-*t* graph, the slope of the tangent at any point equals the particle's velocity at that point.
- The steeper the slope (positive or negative), the greater the particle's speed in the positive or negative *x*-direction.

Acceleration

- Average Acceleration
  - ➢ The average acceleration is the change in velocity divided by the time interval during which the change occurs

$$a_{\text{av-}x} = \frac{\Delta v_x}{\Delta t} = \frac{v_{2x} - v_{1x}}{t_2 - t_1} \ (\text{m/s}^2) \qquad (2.4)$$

- Instantaneous Acceleration
  - ➢ The instantaneous acceleration is the limit of the average acceleration as the time interval approaches zero, or equals the derivative of velocity with time

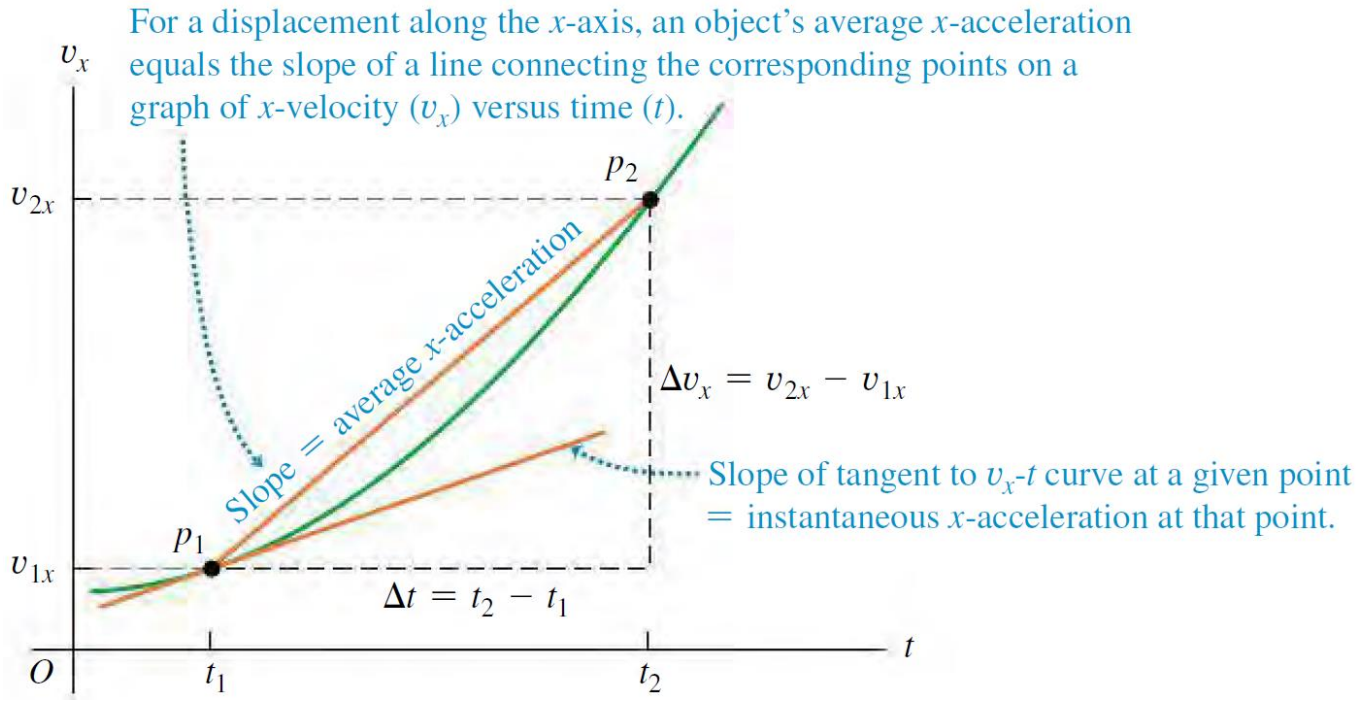
$$a_x = \lim_{\Delta t \to 0} \frac{\Delta v_x}{\Delta t} = \frac{dv_x}{dt} \qquad (2.5)$$

$$v_x = \lim_{\Delta t \to 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt}$$

$$a_x = \frac{dv_x}{dt} = \frac{d}{dt}\left(\frac{dx}{dt}\right) = \frac{d^2x}{dt^2} \qquad (2.6)$$

Acceleration

- Acceleration: from the $v_x(t)$-t graph



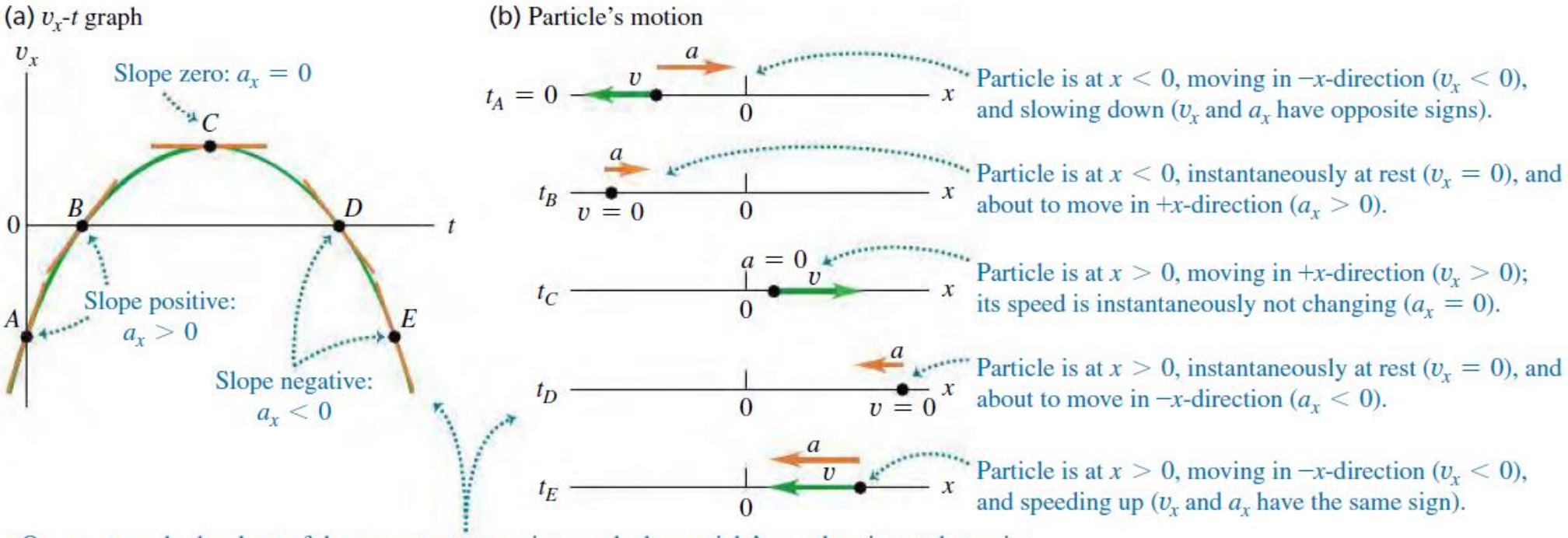
- The instantaneous acceleration at any point is equal to the slope of the tangent to the $v_x(t)$-t curve at that point.

Acceleration

$\begin{cases} a>0 \\ a=0 \\ a<0 \end{cases}$

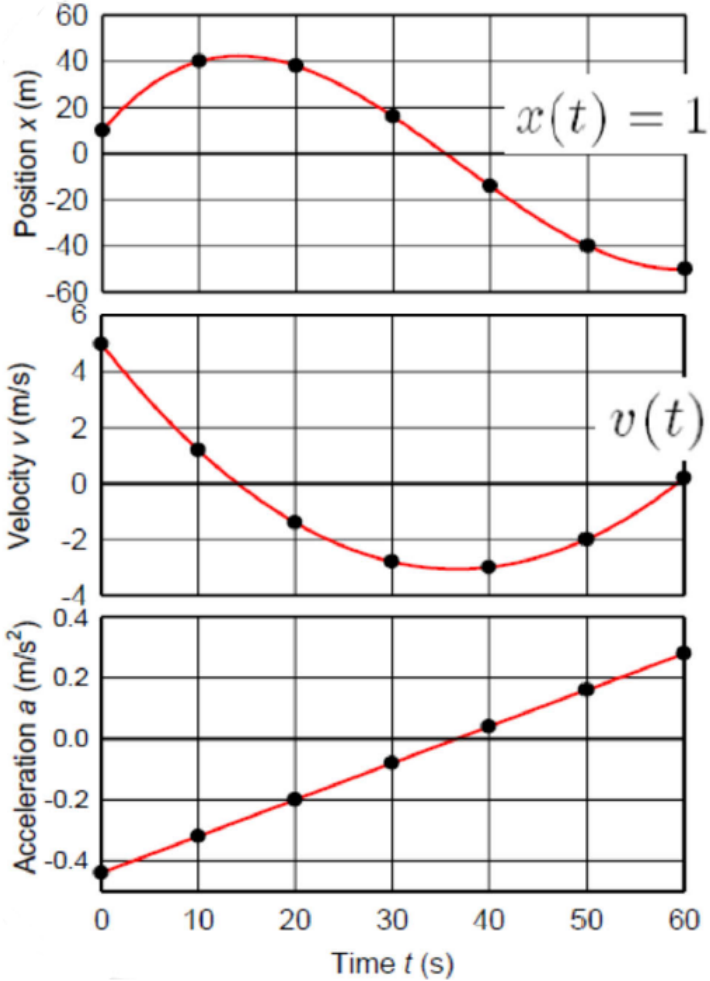
(a) The v_x-t graph of the motion of a another particle.

(b) A motion diagram showing the position, velocity, and acceleration of the particle at each of the times labeled on the v_x-t graph.



- On a v_x-t graph, the slope of the tangent at any point equals the particle's acceleration at that point.
- The steeper the slope (positive or negative), the greater the particle's acceleration in the positive or negative x-direction.

Example for Velocity and Acceleration



$$x(t) = 10 + 5t - 0.22t^2 + 0.002t^3$$

$$v_x = \frac{dx}{dt}$$

$$v(t) = 5 - 0.44t + 0.006t^2$$

$$a_x = \frac{dv_x}{dt}$$

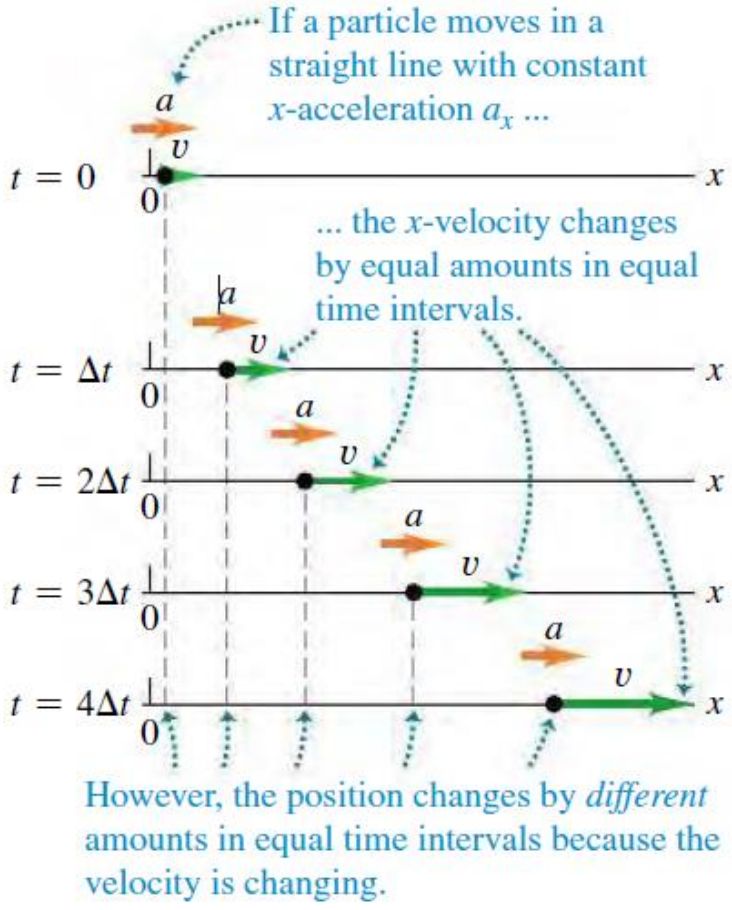
$$a(t) = -0.44 + 0.012t$$

$$\frac{d}{dt}t^n = nt^{n-1}$$

$g = 9.81\ m/s^2$

1D Motion with Constant Acceleration

- The velocity v changes at the **same rate** throughout the motion.
- As an example, a falling body has a constant acceleration if the effects of the air are not important.
- A motion diagram for a particle moving in a straight line in the positive x-direction with constant positive x-acceleration a_x.



1D Motion with Constant Acceleration

- When the x-acceleration a_x is constant, the average x-acceleration $a_{av\text{-}x}$ for any time interval is the same as a_x.
- This makes it easy to derive equations for the position x and the x-velocity v_x as functions of time.
- The expression for v_x can be derived as follows:

$$a_x = \frac{v_{2x} - v_{1x}}{t_2 - t_1} \qquad (2.7)$$

- let $t_1 = 0$ and let t_2 be any later time t. We use the symbol v_{0x} for the initial x-velocity at time $t = 0$; the x-velocity at the later time t is v_x.

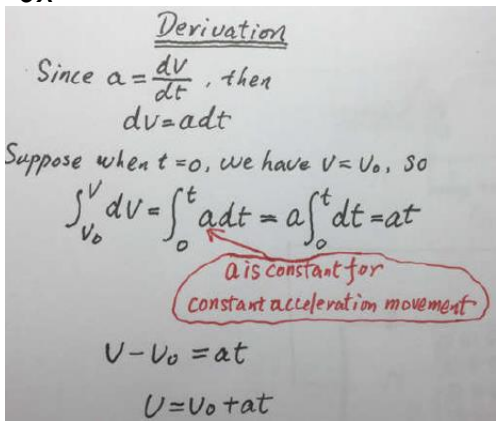
$$a_x = \frac{v_x - v_{0x}}{t - 0}$$

- Finally, we get

x-velocity at time t of a particle with **constant x-acceleration**

$$v_x = v_{0x} + a_x t \qquad (2.8)$$

v_x: x-velocity of the particle at time 0; a_x: Constant x-acceleration of the particle; t: Time



1D Motion with Constant Acceleration

- To derive the equation for the position x as a function of time when the x-acceleration is constant, two different expressions for the average x-velocity $v_{av\text{-}x}$ are used:

➢ $$v_{av\text{-}x} = \frac{x - x_0}{t} \quad (2.9)$$ where $t_0 = 0$

$v_{av\text{-}x}$ for the time interval from 0 to t is simply the average of the x-velocities at the beginning and end of the interval.

➢ $$v_{av\text{-}x} = \tfrac{1}{2}(v_{0x} + v_x) \quad \text{(constant } x\text{-acceleration only)} \quad (2.10)$$

- We know that Eq (2.8) $v_x = v_{0x} + a_x t$, substituting Eq (2.8) to Eq (2.10):

$$v_{av\text{-}x} = \tfrac{1}{2}(v_{0x} + v_{0x} + a_x t) = v_{0x} + \tfrac{1}{2}a_x t \quad (2.11)$$

- Combining Eq 2.11 and Eq 2.9, we get:

$$v_{0x} + \tfrac{1}{2}a_x t = \frac{x - x_0}{t} \quad \text{or}$$

Position of the particle at time 0

Position at time t of a particle with **constant** x**-acceleration**

$$x = x_0 + v_{0x}t + \tfrac{1}{2}a_x t^2 \quad (2.12)$$

Time

x-velocity of the particle at time 0

Constant x-acceleration of the particle

1D Motion with Constant Acceleration

- It's often useful to have a relationship for position, x-velocity, and (constant) x-acceleration that does not involve time.
- To obtain this, we first solve Eq (2.8) for t and then substitute the resulting expression into Eq (2.12):

$$t = \frac{v_x - v_{0x}}{a_x} \qquad x = x_0 + v_{0x}\left(\frac{v_x - v_{0x}}{a_x}\right) + \tfrac{1}{2}a_x\left(\frac{v_x - v_{0x}}{a_x}\right)^2$$

- Transfer the term x_0 to the left side, multiply through by $2a_x$, and simplify:

$$2a_x(x - x_0) = 2v_{0x}v_x - 2{v_{0x}}^2 + v_x^2 - 2v_{0x}v_x + {v_{0x}}^2$$

- Finally,

x-velocity at time t of a particle with **constant x-acceleration**

$$v_x^2 = {v_{0x}}^2 + 2a_x(x - x_0) \qquad (2.13)$$

x-velocity of the particle at time 0

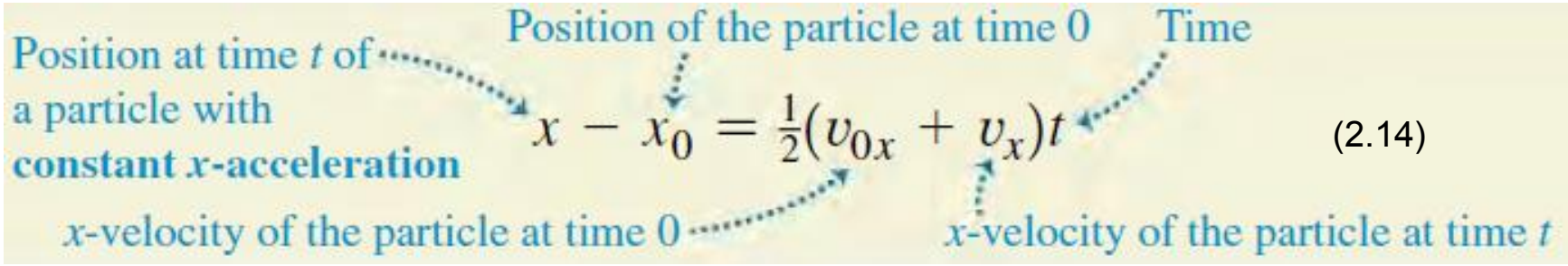
Constant x-acceleration of the particle

Position of the particle at time t

Position of the particle at time 0

1D Motion with Constant Acceleration

- We can get one more useful relationship by equating the two expressions for $v_{av\text{-}x}$, Eqs (2.9) and (2.10), and multiplying through by t:



- Since Eq (2.14) does not contain the x-acceleration a_x, it can be handy when a_x is constant but its value is unknown.

1D Motion with Constant Acceleration

- Important formulae for 1D motion with constant acceleration

$$v = v_0 + at \quad (2.8)$$

$$x = x_0 + v_0 t + \frac{1}{2}at^2 \quad (2.12)$$

$$v^2 = v_0^2 + 2a(x - x_0) \quad (2.13)$$

$$x - x_0 = \frac{1}{2}(v + v_0)t \quad (2.14)$$

where v_0 and x_0 are initial values of the velocity and position.

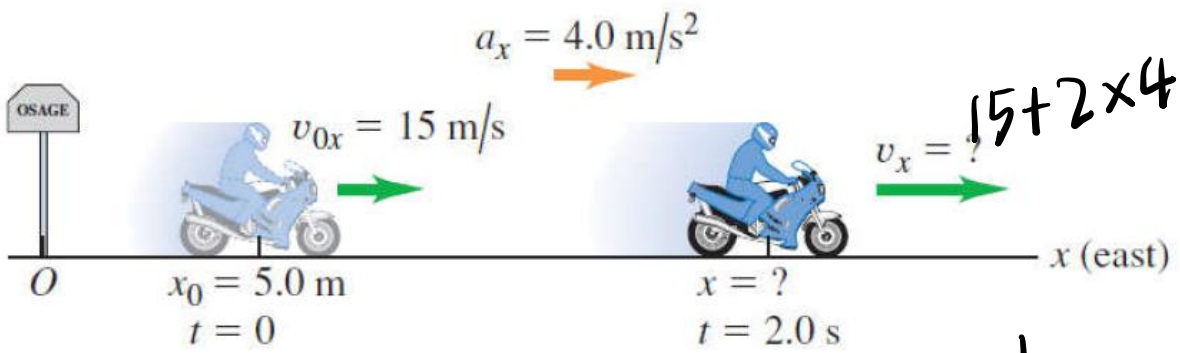
- **Valid only for motion with constant acceleration**

Example for Constant Acceleration

Example 2.4 Constant-acceleration calculations

A motorcyclist heading east through a small town accelerates at a constant 4.0 m/s^2 after he leaves the city limits (Fig. 2.20). At time $t = 0$ he is 5.0 m east of the city-limits signpost, moving east at 15 m/s. (a) Find his position and velocity at $t = 2.0$ s. (b) Where is he when his velocity is 25 m/s?

2.20 A motorcyclist traveling with constant acceleration.



$15+2\times4=23\ m/s$

$x = 15\times2 + \frac{1}{2}\times4\times2\times2 + x_0$

= 43m

Example for Constant Acceleration

SOLUTION (a) Since we know the values of x_0, v_{0x}, and a_x, then we have

$$\begin{aligned} x &= x_0 + v_{0x}t + \tfrac{1}{2}a_x t^2 \\ &= 5.0\ \text{m} + (15\ \text{m/s})(2.0\ \text{s}) + \tfrac{1}{2}(4.0\ \text{m/s}^2)(2.0\ \text{s})^2 \\ &= 43\ \text{m} \end{aligned}$$

$$\begin{aligned} v_x &= v_{0x} + a_x t \\ &= 15\ \text{m/s} + (4.0\ \text{m/s}^2)(2.0\ \text{s}) = 23\ \text{m/s} \end{aligned}$$

(b) From

$$v_x^2 = v_{0x}^2 + 2a_x(x - x_0)$$

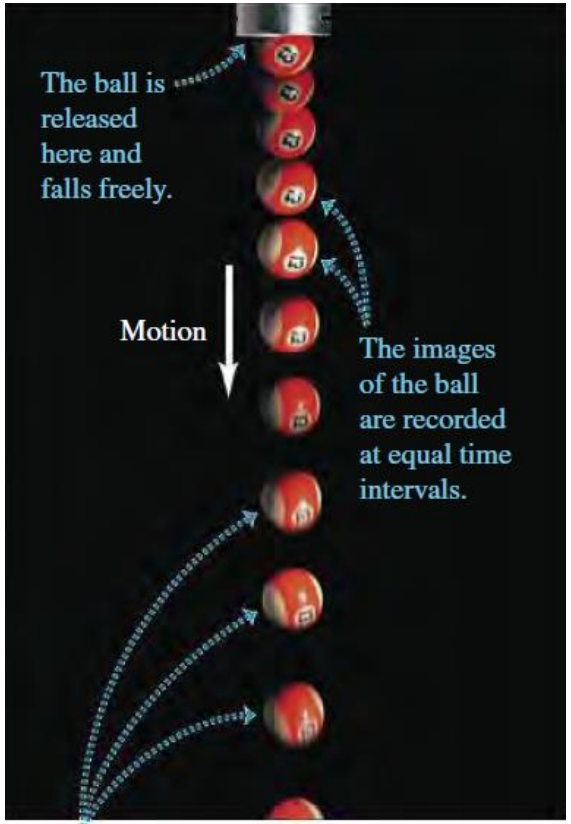
we can get

$$x = x_0 + \frac{v_x^2 - v_{0x}^2}{2a_x}$$

$$= 5.0\ \text{m} + \frac{(25\ \text{m/s})^2 - (15\ \text{m/s})^2}{2(4.0\ \text{m/s}^2)} = 55\ \text{m}$$

Free Falling Objects 自由落体

- Free falling is a typical 1D motion.
- All objects, when dropped, will fall toward the Earth with nearly constant acceleration.
- The free fall acceleration or acceleration due to gravity is given by $g \approx 9.80$ m/s^2, assuming that the
  - height is less than the radius (半径) of the Earth (R_e)
  - air resistance is neglected
- An object in free fall will experience an acceleration downward of **magnitude** g, that is, 1D motion under constant acceleration.



• The average velocity in each time interval is proportional to the distance between images.
• This distance continuously increases, so the ball's velocity is continuously changing; the ball is accelerating downward.

Free Falling Objects

- By convention the vertical direction y is positive upward.
- Hence, the acceleration due to gravity (downward), where

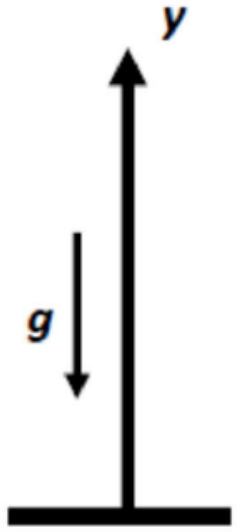
$$a = -g$$

- Corresponding kinematic equations:

$$v = v_0 - gt \qquad (2.8)$$

$$y = y_0 + v_0 t - \frac{1}{2} g t^2 \qquad (2.12)$$

$$v^2 = v_0^2 - 2g(y - y_0) \qquad (2.13)$$



Free Falling Objects

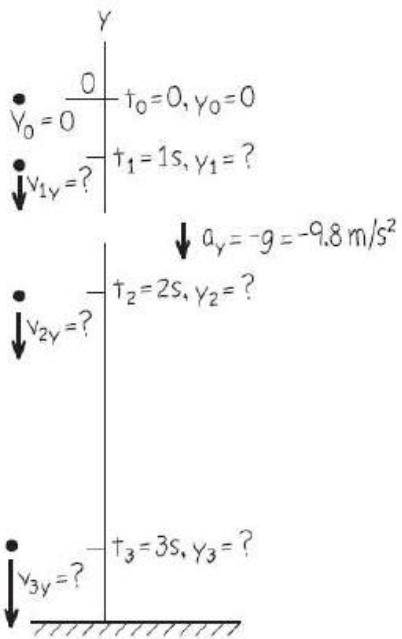
Example 2.6 A one-euro coin is dropped from the Leaning Tower of Pisa and falls freely from rest. What are its position and velocity after 1.0 s, 2.0 s, and 3.0 s?

Solution

The Leaning Tower



Our sketch for the problem



EXECUTE: At a time t after the coin is dropped, its position and y-velocity are

$$y = y_0 + v_{0y}t + \tfrac{1}{2}a_y t^2 = 0 + 0 + \tfrac{1}{2}(-g)t^2 = (-4.9\ \text{m/s}^2)t^2$$

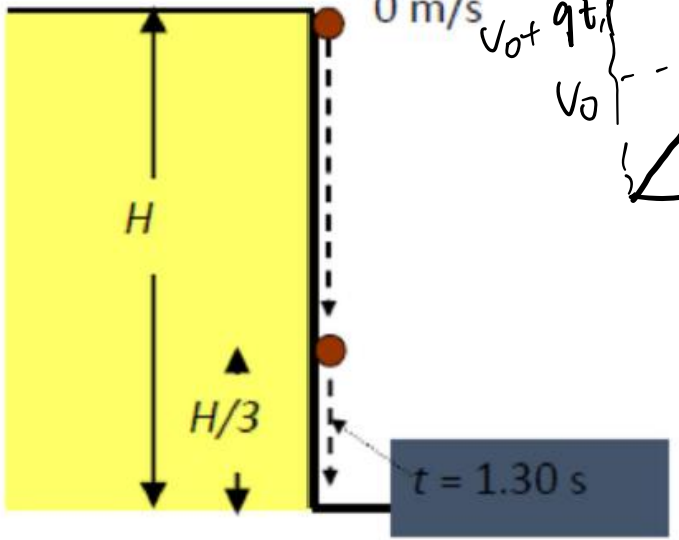
$$v_y = v_{0y} + a_y t = 0 + (-g)t = (-9.8\ \text{m/s}^2)t$$

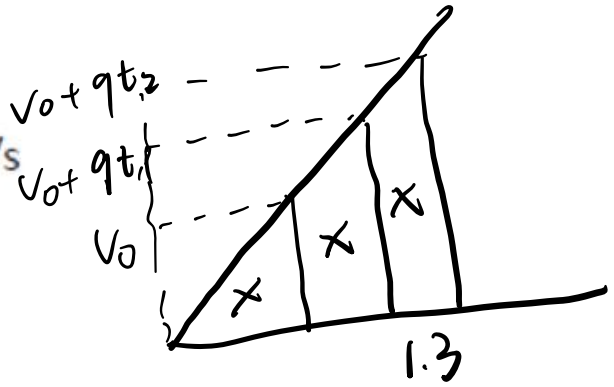
When $t = 1.0$ s, $y = (-4.9\ \text{m/s}^2)(1.0\ \text{s})^2 = -4.9$ m and $v_y = (-9.8\ \text{m/s}^2)(1.0\ \text{s}) = -9.8$ m/s; after 1 s, the coin is 4.9 m below the origin (y is negative) and has a downward velocity (v_y is negative) with magnitude 9.8 m/s.

We can find the positions and y-velocities at 2.0 s and 3.0 s in the same way. The results are $y = -20$ m and $v_y = -20$ m/s at $t = 2.0$ s, and $y = -44$ m and $v_y = -29$ m/s at $t = 3.0$ s.

Free Falling Objects

An alert hiker sees a boulder fall from the top of a distant cliff and notes that it takes $1.30\ \mathrm{s}$ for the boulder to fall from the last third of the way to the ground. What is the height of the cliff? (Ignore air resistance.)





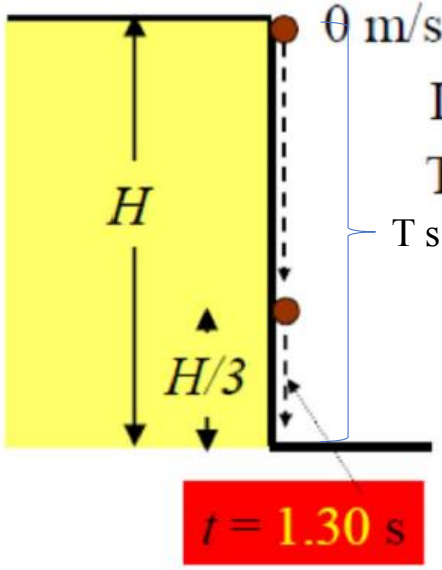
$$x=(v_0+gt_1)\cdot 1.3+\frac{1}{2}g\cdot 1.3^2$$

$$v_0+gt_1+g\times 1.3=v_0+gt_2$$

$$3x=\frac{1}{2}gt^2$$

Free Falling Objects

Solutions



Let T be the time it takes to fall to the ground.

Then:

$$H = \frac{1}{2}gT^2 \quad ; \quad \frac{2}{3}H = \frac{1}{2}g(T-t)^2$$

$$\frac{1}{3}gT^2 = \frac{1}{2}g(T-t)^2 \Rightarrow \frac{2}{3}T^2 = (T-t)^2$$

$$\sqrt{\frac{2}{3}}T = T - t$$

$$T = \frac{t}{1-\sqrt{\frac{2}{3}}} = 7.08 \text{ s}$$

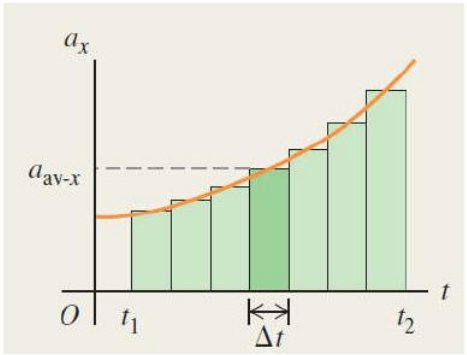
$$H = 246 \text{ m}$$

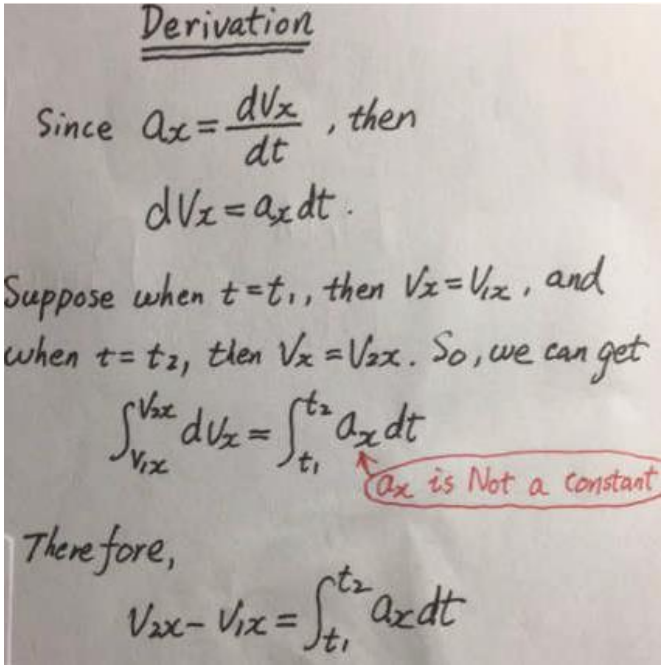
1D Motion with Varying Acceleration

- For 1D motion with a varying acceleration, find the velocity and position as functions of time by integration of the acceleration.

For velocity

$$v_{2x} - v_{1x} = \int_{v_{1x}}^{v_{2x}} dv_x = \int_{t_1}^{t_2} a_x\, dt \qquad (2.15)$$





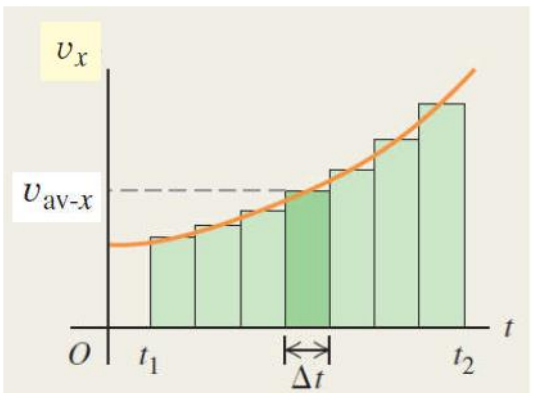
- If $t = 0$, then $v = v_{0x}$. For any later time t, we have

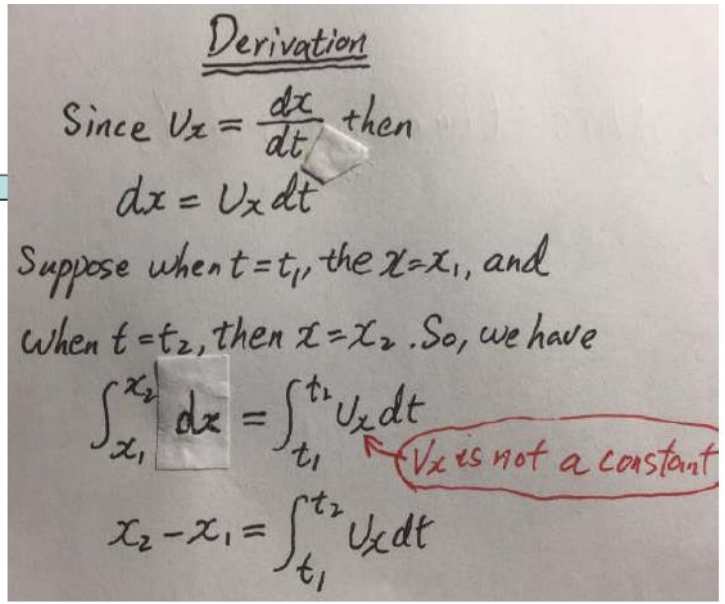
$$v_x = v_{0x} + \int_0^t a_x\, dt \qquad (2.17)$$

1D Motion with Varying Acceleration

For position/displacement

$$x_2 - x_1 = \int_{x_1}^{x_2} dx = \int_{t_1}^{t_2} v_x\, dt \qquad (2.16)$$





- If $t = 0$, then $x = x_0$. For any later time t, we have

$$x = x_0 + \int_0^t v_x\, dt \qquad (2.18)$$

Example for 1D Motion with Varying Acceleration

Example 2.9

Sally is driving along a straight highway in her 1965 Mustang. At $t = 0$, when she is moving at 10 m/s in the positive x-direction, she passes a signpost at $x = 50$ m. Her x-acceleration as a function of time is



$$a_x = 2.0\ \text{m/s}^2 - (0.10\ \text{m/s}^3)t$$

(a) Find her x-velocity v_x and position x as functions of time. (b) When is her x-velocity greatest? (c) What is that maximum x-velocity? (d) Where is the car when it reaches that maximum x-velocity?

Solution (a) At $t = 0$, Sally's position is $x_0 = 50$ m and her x-velocity is $v_{0x} = 10$ m/s.

Then we have

(2.17) $$v_x = v_{0x} + \int_0^t a_x\,dt$$

$$v_x = 10\ \text{m/s} + \int_0^t [2.0\ \text{m/s}^2 - (0.10\ \text{m/s}^3)t]\,dt$$

$$= 10\ \text{m/s} + (2.0\ \text{m/s}^2)t - \tfrac{1}{2}(0.10\ \text{m/s}^3)t^2$$

$$\int t^n\,dt = \frac{1}{n+1}t^{n+1} \quad (n \neq -1)$$

$$\int_0^t t\,dt = \tfrac{1}{2}t^2\Big|_0^t = \tfrac{1}{2}t^2$$

Example for 1D Motion with Varying Acceleration

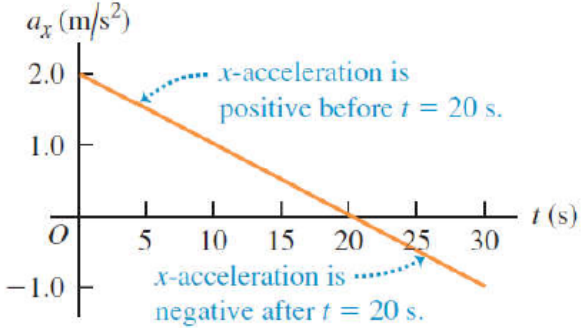
Solution (con't)

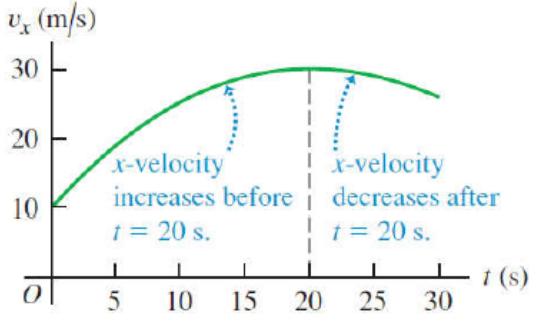
$$x = 50\ \text{m} + \int_0^t [10\ \text{m/s} + (2.0\ \text{m/s}^2)t - \tfrac{1}{2}(0.10\ \text{m/s}^3)t^2]\,dt$$

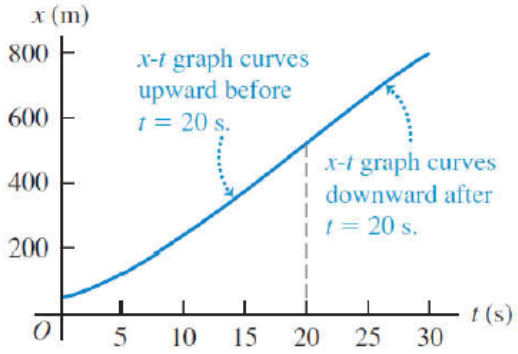
$$= 50\ \text{m} + (10\ \text{m/s})t + \tfrac{1}{2}(2.0\ \text{m/s}^2)t^2 - \tfrac{1}{6}(0.10\ \text{m/s}^3)t^3$$

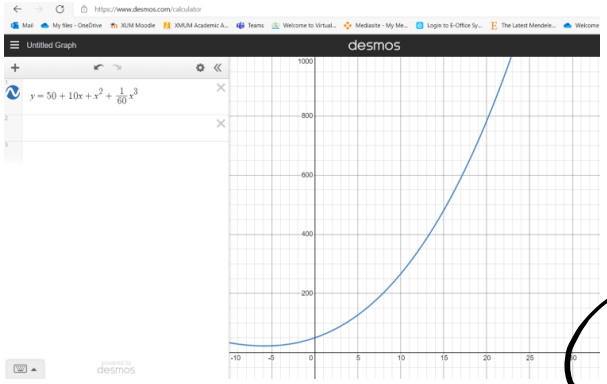
$$x = x_0 + \int_0^t v_x\,dt \qquad (2.18)$$

Then we can plot the curves of a_x-t, v_x-t, and x-t as the above equations. Note that for any time t, the slope of the v_x-t graph equals the value of a_x and the slope of the x-t graph equals the value of v_x.









https://www.desmos.com/calculator

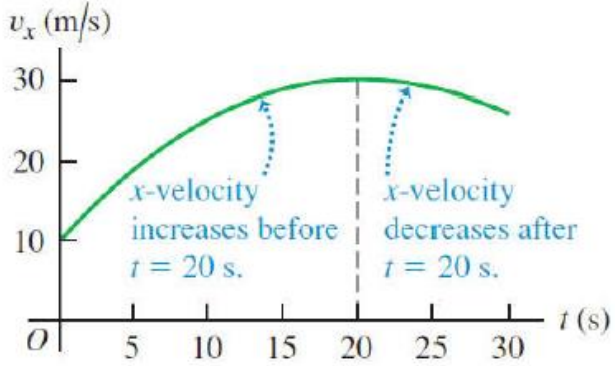
Example for 1D Motion with Varying Acceleration

Solution (con't)

(b) The maximum value of v_x occurs when the x-velocity stops increasing and begins to decrease. At that instant, $dv_x/dt = a_x = 0$. So we set the expression for a_x equal to zero and solve for t:

$$0 = 2.0\ \text{m/s}^2 - (0.10\ \text{m/s}^3)t$$

$$t = \frac{2.0\ \text{m/s}^2}{0.10\ \text{m/s}^3} = 20\ \text{s}$$



(c) We find the maximum x-velocity by substituting $t = 20$ s, the time from part (b) when velocity is maximum, into the equation for v_x from part (a):

$$v_{\text{max-}x} = 10\ \text{m/s} + (2.0\ \text{m/s}^2)(20\ \text{s}) - \tfrac{1}{2}(0.10\ \text{m/s}^3)(20\ \text{s})^2 = 30\ \text{m/s}$$

(d) To find the car's position at the time that we found in part (b), we substitute $t = 20$ s into the expression for x from part (a):

$$x = 50\ \text{m} + (10\ \text{m/s})(20\ \text{s}) + \tfrac{1}{2}(2.0\ \text{m/s}^2)(20\ \text{s})^2 - \tfrac{1}{6}(0.10\ \text{m/s}^3)(20\ \text{s})^3 = 517\ \text{m}$$

Summary

- 1D Motion is discussed and studied with examples.
- Reading materials
  - Chapter 2 Motion in A Straight Line
- Assignment 1
  - Description of coursework is being finalized.
  - Homework on that covers topics from lecture 1 to 5 will be assigned in Week 5, check Moodle announcement for updates.